Welcome to the Year 2095

January 5th, 2095

Dear Aerospace Enterprises,

Welcome to this year's bidding process for the Demeter space settlement in the asteroid belt. The International Deep Space Development Organization (IDSDO) is excited to invite you to participate in this remarkable project.

Established in 2025 and jointly operated by leading aerospace powers worldwide, IDSDO has been dedicated to the exploration, development, and utilization of deep space resources for the benefit of all humanity. Over the past four decades, our achievements in deep space exploration and space resource development have been extraordinary. The innovative space hotels we pioneered in low Earth orbit (LEO) have become standout examples of commercial success. The completed solar power satellite (SPS) network effectively resolves Earth's energy challenges through the utilization of space-based power generation and transmission technology. Our lunar mining bases and industrial space settlements located at Earth-Moon Lagrange points L4 and L5 continue to yield valuable resources, propelling our progress into the vast reaches of space. This era witnessed a flourishing of space industries across sectors such as refining, semiconductors, energy, pharmaceuticals, and tourism. In the 2060s, IDSDO led Venus-focused investment and construction, establishing multiple orbital and atmospheric space settlements. Their food, dry ice, and liquid nitrogen supplies supported inner solar system deep-space exploration. From the 2070s, Mars development became IDSDO's top priority. Following over 20 years of efforts, a systematic deep space infrastructure network was built, including Martian surface industrial and scientific research bases, plus orbital space cities.

Since 2085, after 40 years of lunar and near-Earth asteroid (NEA) mining, commercial mining of asteroid belt became a new hot spot. With the help of new space fusion power generators and engines, a group of commercial companies began to search for targets in the asteroid belt between Mars and Jupiter for experimental commercial mining. The technology of zero/low gravity mining performed by robots was rapidly applied in this area. IDSDO also made a visionary decision, started to lead a series of satellite and unmanned probe missions, and set up an asteroid information database (AID) which was open to all the world .

By 2090, the AID had collected nearly 8,000 valuable pieces of information of asteroids, including their orbital information, spin state, radius, mass, composition and vein distribution. At least five commercial companies became successful in this area. Their usual mining model was used to build a 10-20 crew-manned, fusion-powered mining and smelting base in the solar orbit near the target asteroid. The small unmanned mining robots were then dropped onto the target asteroid and worked on the detected vein. Then, the ore was transported to the base by electromagnetic mass driver or unmanned small spacecrafts. At the base, the ore was refined or smelted according to its type to produce various valuable materials such as aluminum, magnesium, gold, platinum, water, liquid oxygen, and high-purity silicon. When the resources of the target asteroid were exhausted, most of the equipment would be recovered and the mining base towed by the spacecraft to the next target asteroid to continue operations.

In 2095, to provide better services for asteroid mining companies and improve the efficiency of resource acquisition, IDSDO has decided to build a large space settlement in the asteroid belt. This space city will be constructed in a solar orbit near the Psyche. Thanks to the Psyche's abundant metal deposits, the area has become a hub for asteroid mining. Building a large-scale infrastructure here will undoubtedly greatly advance the development of the asteroid belt. The settlement will mainly provide re-fueling, resupplies, maintenance and rescue services for mining bases and cargo spacecraft. Raw materials, high-purity silicon, fuel, etc. transported from the mining site will be traded and/or processed in it and then transported to other places. Crews from mining bases will also visit this new city for recreational and medical services. IDSDO has selected "Demeter" as the name of this new settlement, after the goddess of harvest in Greek mythology, also recognizing the first main-belt asteroid discovered, Ceres, her another name in Roman mythology. The name Demeter also indicates that asteroid belt will not only serve as a resource base for the inner solar system, but also provide a constant resupply of food and fuel for humans to travel to deeper space.IDSDO will officially release the Request for Proposal (RFP) containing detailed requirements. Your company is invited to join us in this noble endeavor by designing and submitting a proposal that fulfills all the specified requirements. We wish you the best of luck in this exciting challenge in the days ahead.

Dr. Richard Doe Richard Doe

Chair of International Deep Space Development Organization